C^{3}DG: Conditional Domain Generalization for Hyperspectral Imagery Classification With Convergence and Constrained-Risk Theories

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***Abstract*—Hyperspectral imagery (HSI) classification faces the challenge of hyperspectral-monospectra, where distinct classes exhibit similar spectral signatures. The spatial–spectral approach is a common solution but often inflates accuracy due to test pixels overlapping with training patches. While domain generalization shows promise in addressing hyperspectral-monospectra, it still struggles to distinguish spectrally similar samples from different domains. In this work, we address hyperspectral-monospectra using spectral information alone and propose a convergence- and error-constrained conditional domain generalization (C^{3}DG) method for HSI classification. The main contributions of this work are twofold: 1) the conditional revising inference block (CRIB) and 2) the theoretical guarantees on model convergence and generalization error. CRIB forms the core of our method, employing a shared encoder and multibranch decoders to fully leverage conditional label distributions during training. This design achieves structural decoupling aligned with the generation mechanisms of HSI. To ensure stable optimization and bounded generalization error, we establish an optimization convergence theorem and a risk upper bound theorem. The convergence theorem shows that the gradients of the loss components consistently align, ensuring stable model optimization. The risk bound theorem establishes a concrete error bound by connecting test-time training (TTT) with recent developments in sharpness-aware minimization. Experiments on three benchmark datasets demonstrate the superior performance of C^{3}DG. The code is available at `github.com/Lab-PANbin/C3DG`**

***Index Terms*—Conditional domain generalization, hyperspectral classification, hyperspectral-monospectra.**

I. Introduction

HYPERSPECTRAL imagery (HSI) captures detailed spectral information across a broad range of wavelengths, enabling precise material classification. As a result, HSI classification plays a vital role in diverse applications such as agricultural monitoring, environmental observation, and mineral exploration [1], [2], [3]. However, a major challenge in HSI classification is the phenomenon of *hyperspectral-monospectra* [4], [5], [6], where identical spectra are reflected by different classes within hyperspectral datasets. This phenomenon substantially degrades classification accuracy and limits the effectiveness of standard models. Specifically, hyperspectral-monospectra differs from general spectral ambiguity or class overlap in that it arises from the physical imaging process: distinct ground materials under different environmental conditions (e.g., illumination, sensor characteristics) can produce nearly identical spectral signatures. This domain-induced spectral degeneracy is a dominant failure mode in cross-scene benchmarks, as evidenced by the significant accuracy drops observed when models trained on one scene are directly applied to another (see Section IV for quantitative analysis).

Recent hyperspectral classification methods integrate spatial features to mitigate the effects of hyperspectral-monospectra. Common approaches include autoencoder-based encoder–decoder networks [7], [8], [9], which improve feature extraction. Unsupervised and semi-supervised classification methods [10], [11], [12], [13], [14], [15], [16], including prototype-based information compensation networks [15] and nearest neighbor-based contrastive learning approaches [16], leverage abundant unlabeled samples to enhance classification accuracy, and band selection or reduction techniques [17], [18], [19], [20], [21], which filter out the most informative bands for classification. Structurally, convolutional neural networks (CNNs) designed for HSI classification [22], [23] effectively exploit both spatial and spectral features. Graph-based methods [24], [25], [26], [27] adopt a global–local perspective to model correlations among both adjacent and distant pixels. Transformer-based models [28], [29], [30], [31], [32], [33] divide HSI into spatial–spectral patches, enabling joint exploitation of spatial and spectral information. In particular, deep symmetric fusion transformers [33] have demonstrated strong performance in multimodal remote sensing data classification. Mamba-based models [34], [35], [36], [37], [38], [39], tailored for efficient long-range sequence modeling, treat spectral bands as a 1-D ordered sequence. Recent advances such as LSFMamba [39] further enhance multimodal land cover classification through local-enhanced spiral fusion mechanisms. Additionally, hyperspectral image super-resolution methods based on unsupervised reflectance migration [40] and diffusion-driven mutual enhancement [41] have shown promising results in improving spectral fidelity, while CNN-VMamba synergistic

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networks [42] have advanced remote sensing image change captioning.

As noted in prior studies [43], [44], incorporating spatial features may artificially inflate accuracy when test samples overlap with the training set. Moreover, the hyperspectral-monospectra problem persists even when spatial information is considered [6]. For instance, cement roads and rooftops at different elevations may exhibit indistinguishable spatial–spectral signatures. Additionally, spatial–spectral algorithms degrade significantly when only spectral information is available [44], [45], even in the era of foundation models [46]. Consequently, spectral-only approaches have gained increasing attention in recent research.

Domain generalization [47] offers a promising solution to the hyperspectral-monospectra problem. These methods [48], [49], [50], [51] extract domain-specific features, enabling classifiers to distinguish between visually similar samples from different domains, thereby producing distinct classification outcomes. Specifically, test-time training (TTT) methods [52], [53] adapt the model during inference using a context network, enabling distinct outputs for samples with similar spectral magnitudes but from different domains.

Recent studies on domain generalization in HSI classification primarily focus on fixed-output approaches, which fall into two main categories. Most existing methods [54], [55], [56] adopt single-source domain generalization, leveraging domain-generative data augmentation to learn domain-invariant features. Multisource methods [57], [58] manually define source domains, aiming to minimize cross-domain discrepancies while preserving intraclass diversity to enhance generalization. However, these approaches still rely on spatial information to mitigate the hyperspectral-monospectra problem. Two major challenges remain as follows.

1) As noted earlier, these methods generate fixed outputs and are not explicitly designed to address the hyperspectral-monospectra problem.
2) Most existing methods emphasize technical implementation while neglecting theoretical foundations, resulting in a lack of convergence analysis and generalization error bounds. This may introduce conflicts among components and hinder optimization.

To address these challenges, it is essential to design a framework that not only adapts predictions conditionally based on domain characteristics, but also provides theoretical guarantees to ensure stable training and bounded generalization error. The motivation for combining convergence constraints with conditional domain generalization is threefold: 1) conditional branching enables the model to exploit class-specific distributional information, which is critical for resolving hyperspectral-monospectra; 2) without convergence guarantees, the multiobjective optimization involving classification, reconstruction, and reversibility losses may lead to conflicting gradients and unstable training; and 3) a bounded generalization risk ensures that the test-time adaptation mechanism does not degrade performance on unseen domains. These considerations jointly motivate the design of our convergence- and error-constrained conditional domain generalization (C^3DG) framework.

Therefore, we propose C^3DG for HSI classification, which generates distinct outputs for spectrally similar samples with different domain characteristics to address the hyperspectral-monospectra problem. Inspired by TTT, we introduce the conditional revising inference block (CRIB) and present theoretical guarantees on training convergence and generalization error bounds. CRIB consists of a shared decoupled encoder and a multibranch, domain-aware decoder to achieve identifiable disentanglement and fully exploit the conditional input distribution (i.e., label information) during training. For convergence, we prove that the inner product between gradients of different loss terms remains consistently positive, ensuring optimization stability. To derive the generalization bound, we establish a connection between TTT and recent sharpness-aware minimization theories, constructing a constrained risk bound.

1) We propose a conditional domain generalization strategy for HSI classification that addresses the hyperspectral-monospectra problem by adapting predictions during inference.
2) We design a multibranch CRIB to effectively leverage the conditional distribution of input samples.
3) We provide theoretical guarantees for the proposed method, including a convergence theorem and an upper bound on the classification risk.

II. Methodology

A. Preliminaries

Most models encounter significant accuracy drops when applied to unseen target domains after being trained on source domains. Consequently, domain generalization, which aims to enhance a model's generalization capability from source domains to other unseen domains, has attracted academic attention. A brief definition can be summarized as follows.

Domain Generalization: Given M source domains with labels $\{X_i\}_{i=1}^{M}$, $X_i = \{x_j, y_j\}_{j=1}^{N_i}$. N_i is the number of samples in domain X_i. x_j, y_j are the data and labels. The aim is to obtain a generalizable model for the unseen target domains with $\{X_i\}_{i=1}^{M}$, formally

$$\min \mathbb{E}_{D\in\mathcal{D}}\left[\mathbb{E}_{x\in D}\left[\mathcal{R}\left(f\left(x\right), y\right)\right]\right] \tag{1}$$

where $\mathcal{D}$ is the total domain space and $\mathcal{R}$ is risk function.

TTT methods of domain generalization aim to learn models that adapt at testing to domain shift using unlabeled test data, which can be formally defined as follows.

Test Time Training: Aims to learn a function shift from the input batch $\mathcal{B}$ and current model parameters θ to the C^3DG network parameter θ', i.e., $f_{shift} : \mathcal{B}, \theta \rightarrow \theta'$, that satisfy the following optimization problem:

$$\min_{\theta,\phi} \hat{\mathcal{E}}\left(\theta, \phi\right) = \mathbb{E}_{p_z}\left[\mathbb{E}_{p_{xy|z}}\left[\frac{1}{K}\sum_{k=1}^{K}\mathcal{R}\left(f\left(x_k;\theta'\right), y_k\right)\right]\right] \tag{2}$$

where $\theta' = f_{shift}(\theta, x_1, \ldots, x_K; \phi)$ and ϕ the parameters of f_{shift}.

An indirectly but workable implementation approach [52] for f_{shift} is as follows:

$$f\left(x, \theta'\right) \simeq f\left(\text{concat}\left(x, x_{context}\right)\right) \tag{3}$$

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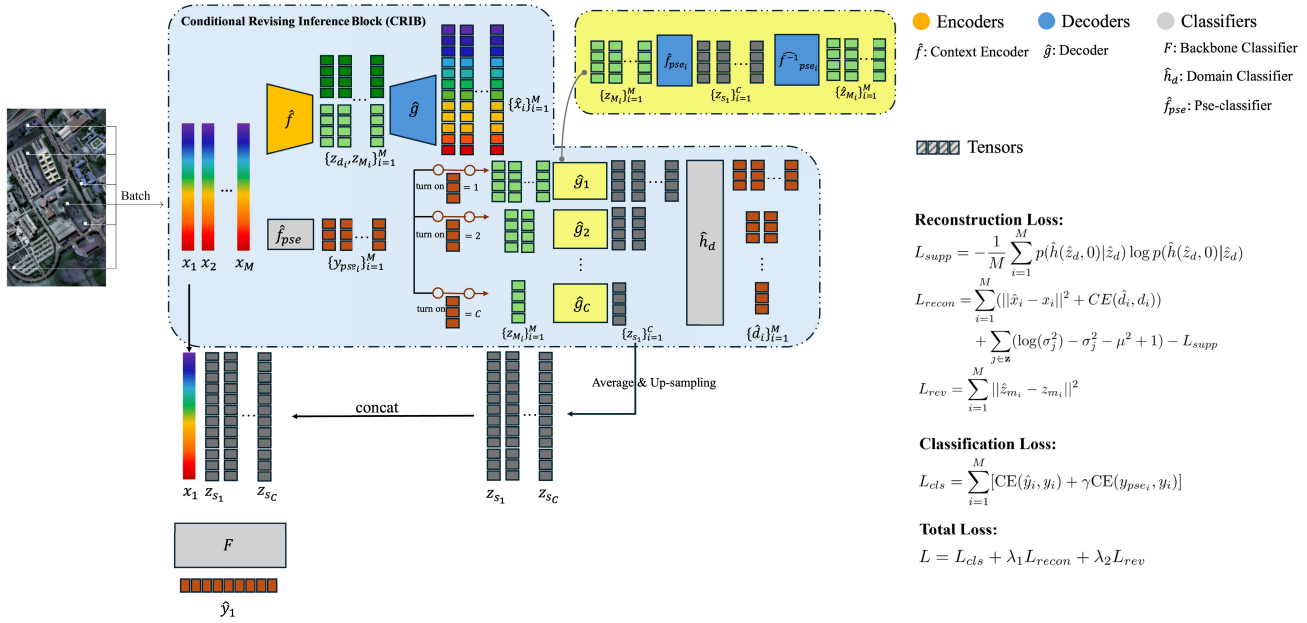


Fig. 1. Framework of the proposed C^{3}DG architecture is structured as follows: The context encoder in the CRIB comprises C branches, each processing distinct conditional distributions while sharing the same encoder. Specifically, C forward networks serve as decoders, generating the final revision features. During inference, each pixel is assigned to a specific branch for revision feature extraction, determined by the pseudoclassifier's classification. The revision features are then concatenated with the original features and processed by the classifier.

where $x_{context} = f_{cont}(x)$ is the context feature and f_{cont} is the context network. These settings regard the combining of the features with test data as the shift in the model parameters, as summarized in Algorithm 1.

Algorithm 1 Training Stage of C^{3}DG

Input: M source domains with labels $\{X_i\}_{i=1}^{M}$, $X_i = \{x_j, y_j\}_{j=1}^{N_i}$
Output: The parameters $\theta_D, \theta_F, \theta_f$ of C^{3}DG D, backbone predictor F and pseudoclassifier f_{pse}.

Initialize: $\theta_G, \theta_F, \theta_f$
for *epoch* = 1 to T **do**
  Predict pseudo auxiliary labels
  $\hat{y}_{i_{pse}} = f_{pse}(x_i)$
  Extract Domain Context Features
  $(\hat{z}_{d_i}, \hat{z}_{m_i}) = \hat{f}(x_i)$
  $\hat{z}_{s_i} = \hat{f}_{y_i}(\hat{z}_{m_i})$
  for c = 1 to C **do**
    $\mathbf{z}_{s_c} = \mathrm{BI}(\underset{i\leq N}{\mathrm{mean}}\{\hat{z}_{s_i} | y_i = c\})$ # BI: Bilinear Interpolate
    $\mathbf{x} = \mathrm{concat}(\mathbf{x}, \mathbf{z}_{s_c})$
  end for
  Predict the Labels
  $\hat{\mathbf{y}} = F(\mathbf{x})$
  Calculate the total loss $L = L_{cls} + \lambda_1 L_{recon} + \lambda_2 L_{rev}$ via Eq. (9), (8), (5)
  // Note: L_{cls}, L_{recon}, and L_{rev} are computed independently and can be parallelized
  Update the model by gradient descent
end for

B. Overall Framework

In this section, we formally introduce the C^{3}DG framework. Our method can be integrated with any hyperspectral classification approach as a baseline by changing the Backbone Classifier F in the flowchart Fig. 1. For any classifier F, our method serves as a domain feature extractor through an auxiliary branch of CRIB architecture, thereby being able to provide revised outputs for similar inputs but with different domain information. Specifically, we employ a simple 1D-CNN network, of which the extractor is of three 1dConv-layers and the classifier is a two-linear layer structure ending with a softmax, as the baseline for convenience.

As illustrated in Fig. 1, spectra from the input batch of different pixels (denoted as $x_1, x_2 \ldots x_M$) are firstly fed into the CRIB branch to extract the domain feature of these spectra. Specifically, the context encoder $\hat{f}$ in the CRIB processes this batch of data points $\mathcal{B}$ unclasswisedly to obtain the domain variables z_d and the mixed variables z_m. In implementation, all CRIB-related mappings are two-layer MLPs, including the shared encoder $\hat{f}$, branch-specific mapping $\hat{f}_c$, inverse mapping $\hat{f}_c^{-1}$, and pseudolabel classifier $\hat{f}_{pse}$. A sample spectrum x_i in the batch is processed based on its rough pseudolabel y_{pse_i}; that is, only the forward encoder $\hat{f}_{pse_i}$ of the y_{pse_i}-th branch is turned on to process x_i to extract the class-independent variable z_{s_i}, which is then sorted into discrete sets $\mathcal{Z}_{s_1} = \{z_{s_i} \mid y_{pse_i} = 1\}, \ldots, \mathcal{Z}_{s_C} = \{z_{s_i} \mid y_{pse_i} = C\}$ (C is the total class number). The output sets of all branches $\mathcal{Z}_{s_1}, \ldots, \mathcal{Z}_{s_C}$ are taken means set-wisely and then reshaped to match the size of the input spectra point. Consequently, the average reshaped $z_{s_1}, \ldots, z_{s_C}$ are concatenated with $x_1, x_2 \ldots x_M$ channel-wisely to form the final revision $\{\mathrm{Concat}(x_1, z_{s_1}, \ldots, z_{s_C})\}_{i=1}^{M}$ before feeding into the backbone classifier F to obtain the final outputs $\{\hat{y}_i\}_{i=1}^{M}$. Particularly, the real class labels are utilized for deciding the CRIB branches during the training phase.

C. Conditional Revising Inference Block

1) Architectures of CRIB: A straightforward approach to characterize $\{P(X|c)\}_{c=1}^{C}$ is developing C branches of individual

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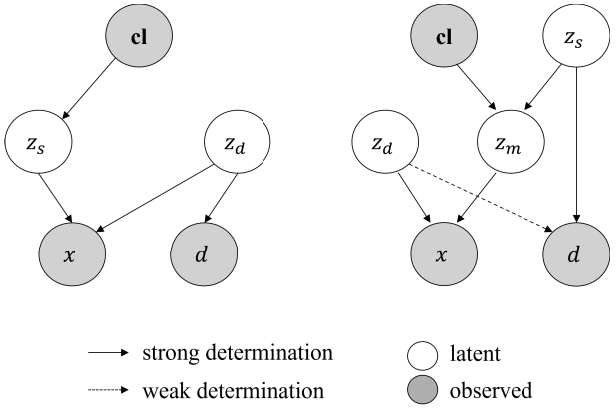


Fig. 2. Causal map of HSI generating. (Left) From previous work and (right) is our finer considerated map. The left casual map assume that the disentanglement of domain-specific and invariant feature is complete, which is in feasible in practice. Therefore, we adopt the right paradigm that impose fewer restrictions on the disentanglement.

context networks tailored for each different conditional distribution. However, this implementation significantly enlarges the model scale since the complexity of context networks is comparably large and proportional to the usually extensive number of classes C. It is detrimental, not only augmenting the computational demands for training but also leading to a pronounced imbalance between the main classification and auxiliary tasks. This imbalance contradicts an empirical observation [52] that the auxiliary task should be subordinate and not undermine the main task.

To address the parameter redundancy arising from the multiple VAE branches, our proposed model ensures that most parameters are shared across these branches. Additionally, it utilizes divergent decoders, which allow for the production of distinct context features tailored to each class. This approach effectively balances the need for specificity in the model's outputs with the efficiency of parameter usage.

Why Reversible-Inspired Architecture Instead of Flow-Based Alternatives: We adopt a reversible-inspired dual-network structure for the decoder rather than fully flow-based (e.g., normalizing flow) architectures for the following reasons. First, flow-based models impose strict architectural constraints (e.g., invertible layers with tractable Jacobian determinants), which significantly limit the expressiveness of each decoder branch. In contrast, our reversible-inspired design relaxes these constraints by training a pair of forward and inverse networks ($\hat{f}_c$ and $\hat{f}_c^{-1}$) with a reversibility loss L_{rev}, allowing more flexible feature transformations. Second, flow-based architectures typically require substantially more parameters and computational resources, which conflicts with our goal of keeping the auxiliary task lightweight relative to the main classification task. Third, our approach naturally integrates with the VAE-based encoder structure and the causal disentanglement framework depicted in Fig. 2, where the decoder needs to separate z_m into z_s conditioned on class labels—a task that does not require exact density estimation but benefits from approximate invertibility.

As depicted in Fig 2, we formalize the inherent generative mechanism of generating the hyperspectral pixels through a two-stage causal diagram. First, the class-independent variable z_s and the label factor c jointly determine the mixed variable z_m which is crucial for label classification. Subsequently, the domain variable z_d, which could be factors such as sensor models or illumination intensity that cause inevitable hyperspectral-monospectra and are challenging to characterize, generated the real-life spectra merged with the mixed variable z_m. The comprehensive characterization z_d, z_s of conditional distributions naturally suits the testing revision for the model's test stage. The essence of addressing hyperspectral-monospectra lies in the successful disentanglement of the input x.

Therefore, the CRIB, based on domain identification variational autoencoder, effectively extracts the feature z_d. As the causality map demonstrates, the ideal context networks for characterizing $\{P(X|c)\}_{c=1}^{C}$ share tremendous portions of their structures. The genuinely requisite architectures to be class-wisely customized is primarily the decoder (denoted as $\hat{f}_c$) that disentangles z_m into z_s.

Algorithm 2 Test Stage of C^{3}DG

Input: A batch of $\{x_i\}_{i=1}^{M}$
Output: Predict labels $\{\hat{y}_i\}_{i=1}^{M}$.

Predict pseudo auxiliary labels
$\hat{y}_{i_{pse}} = f(x_i)$
Extract Domain Context Features
$(\hat{z}_{d_i}, \hat{z}_{m_i}) = \hat{f}(x_i)$
$\hat{z}_{s_i} = \hat{f}_{\hat{y}_{i_{pse}}}(\hat{z}_{m_i})$
for $c =$ `1 to` C **do**
  $\mathbf{z}_{s_c} = \underset{i \le N}{\text{mean}}\{\hat{z}_{s_i} | \hat{y}_{i_{pse}} = c\}$
  $\mathbf{x} = \text{concat}(\mathbf{x}, \mathbf{z}_{s_c})$
end for
Predict the Labels
$\hat{\mathbf{y}} = F(\mathbf{x})$

Our CRIB based on VAE structure accomplishes this disentanglement by utilizing domain labels in a classification task. Specifically, as illustrated in Fig. 1, pixels are initially processed through the encoder $f_{\mu,\sigma}$ to obtain preliminary z_d, z_m. Then the $\{z_m\}$ are processed by different $\hat{f}_c$ depend on their labels. Ultimately, the fully customized domain information z_d, z_s is employed as the context feature z, as summarized in Algorithm 2.

2) Optimization for CRIB: For the training of the public encoder and various decoders, a reconstruction ELBO loss is required both theoretically [59] and intuitively justified. We adhere to the reconstruction loss L_{recon} as part of the auxiliary task. This reconstruction loss ensures the extracting ability of the CRIB while mitigating the risk of overfitting. It is notable that $\hat{f}_u$ is supposed to be reversed in our settings, therefore a reversibility guarantee loss L_{rev} is proposed with a pair of networks $\hat{f}_u, \hat{f}_u^{-1}$. We illustrate the shared decoder as g, the entire classification model as F, and the number of training points as N. β is a hyperparameter. The following are the loss functions:

$$L_{recon} = \sum_{i=1}^{N} \left(\left| \left| g\left(z_{d_i}, z_{m_i}\right) - x \right| \right|^2 + CE\left(\hat{h}_d\left(z_{d_i}, z_{s_i}\right), d_i\right) \right) + \beta \text{KL}\left(\left(z_m, z_s\right), N\left(0, 1\right)\right) \quad (4)$$

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$$L_{rev} = \sum_{i=1}^{N} \left|\left| \hat{f}_{y_i} \left(\hat{f}_{y_i}^{-1} \left(z_{m_i} \right) \right) - z_{m_i} \right|\right|^2 \tag{5}$$

where KL is the KL divergence.

As discussed in [60], we reinforce the disentanglement by mutual information suppression that minimizes $H(\hat{d}|z_d)$. The definition is as follows:

$$H\left(\hat{d}|z_d\right) = -\sum_{\hat{z}_d,\hat{d}} p\left(\hat{z}_d,\hat{d}\right) \log \frac{p\left(\hat{z}_d,\hat{d}\right)}{p\left(\hat{z}_d\right)}. \tag{6}$$

However, for the challenge of analyzing $p(\hat{z}_d,\hat{d})$, we adopt the following approximation to simulate $H(\hat{d}|z_d)$ and detailed theoretical support for the approximation is in Section II-D:

$$\begin{aligned} L_{supp} &= \hat{H}\left(\hat{d}|\hat{z}_d\right): \\ &= -\frac{1}{N}\sum_{i=1}^{N} p\left(\hat{h}\left(\hat{z}_d,0\right)|\hat{z}_d\right) \log p\left(\hat{h}\left(\hat{z}_d,0\right)|\hat{z}_d\right). \end{aligned} \tag{7}$$

By utilizing the closed solution of the KL divergence, the reconstruction loss L_{recon} can be simplified as follows:

$$\begin{aligned} L_{recon} = &\sum_{i=1}^{N} \left(\|g\left(z_{d_i}, z_{m_i}\right) - x\|^2 + CE\left(\hat{h}_d\left(z_{d_i}, z_{s_i}\right), d_i\right)\right) \\ &+ \sum_{j\in\mathbf{z}} \left(\log\left(\sigma_j^2\right) - \sigma_j^2 - \mu^2 + 1\right) - L_{supp}. \end{aligned} \tag{8}$$

The classification loss is as follows, where $CE(\cdot)$ denotes the cross-entropy loss and γ is a hyperparameter

$$L_{cls} = \sum_{i=1}^{N} \left[\mathrm{CE}\left(F\left(x_i\right), y_i\right) + \gamma \mathrm{CE}\left(f_{pse}\left(x_i\right), y_i\right)\right]. \tag{9}$$

From this, we derive the method's loss function

$$L = L_{cls} + \lambda_1 L_{recon} + \lambda_2 L_{rev} \tag{10}$$

where λ_1, λ_2 are the hyperparameters.

D. Theoretical Analysis for Convergence

In this section, we first provide a specific definition of the hyperspectral-monospectra phenomenon from the perspectives of domain generalization and distribution. Then we present a theorem that ensures the convergence of the optimization.

Definition 1 (Hyperspectral-Monospectra): In hyperspectral images, pixels $x_1 = x_2$ from different domains D_1, D_2 but belong to different y_1, y_2 labels. Such an issue is known as the hyperspectral-monospectra phenomenon.

We will first provide an unbiased estimate of the aforementioned entropy loss term H to enable computability in practice.

Property 1: The empirical expectation approximate

$$-\frac{1}{N}\sum_{i=1}^{N} \mathbb{E}_{z_m\sim N(0,I)} p\left(h\left(\hat{z}_{d_i}, z_m\right)|\hat{z}_i\right) \log p\left(h\left(\hat{z}_{d_i}, z_m\right)|\hat{z}_i\right) \tag{11}$$

is unbiased estimation for $H(\hat{d}|z_d)$.

Proof: Sketch 1

$$\begin{aligned} H\left(\hat{d}|\hat{z}_d\right) &= -\frac{1}{N}\sum_{\hat{z}_d,\hat{d}} p\left(\hat{z}_d,\hat{d}\right) \log \frac{p\left(\hat{z}_d,\hat{d}\right)}{p\left(\hat{z}_d\right)} \\ &= -\frac{1}{N}\sum_{\hat{z}_d,z_m} p\left(\hat{z}_d, \hat{h}\left(\hat{z}_d, z_m\right)\right) \log \frac{p\left(\hat{z}_d, \hat{h}\left(\hat{z}_d, z_m\right)\right)}{p\left(\hat{z}_d\right)} \\ &= -\frac{1}{N}\sum_{i=1}^{N} \mathbb{E}_{z_m\sim N(0,I)} p\left(h\left(\hat{z}_{d_i}, z_m\right)|\hat{z}_i\right) \\ &\quad \cdot \log p\left(h\left(\hat{z}_{d_i}, z_m\right)|\hat{z}_i\right). \end{aligned} \tag{12}$$

We sample $z_m = 0$ for computational convenience, thereby

$$\begin{aligned} H\left(\hat{d}|\hat{z}_d\right) &\approx -\frac{1}{N}\sum_{i=1}^{N} p\left(\hat{h}\left(\hat{z}_d,0\right)|\hat{z}_d\right) \log p\left(\hat{h}\left(\hat{z}_d,0\right)|\hat{z}_d\right) \\ &= \hat{H}\left(\hat{d}|\hat{z}_d\right). \end{aligned} \tag{13}$$

Theorem 1 (The Optimization Convergence Theorem): The backward gradient mutual information suppress loss is not contradict to the gradient of the backbone predictor, i.e., $\left\langle \left(\partial \sum CE\left(\hat{h}_d\left(z_d, z_s\right), d_i\right)\right)/\partial w_j, \left(\partial \sum CE\left(\hat{h}_d\left(z_d, z_s\right), d_i\right)\right)/\partial w_j - \left(\partial \sum \hat{d}_{d_i} \log \hat{d}_{d_i}\right)/\partial w_j \right\rangle \geq 0$,, where w_j is any parameter of the backbone predictor in C³DG. Therefore, C³DG leads to a convergence for the disentanglement.

Remark on the Proof Simplifications: The convergence proof below considers a two-class classification setting with a single-layer classifier for analytical tractability. While these simplifications do not directly reflect the multiclass, multilayer network used in practice, they capture the essential gradient alignment property: the mutual information suppression loss does not conflict with the classification objective. This property holds more generally because the core argument relies on the monotonicity of the cross-entropy gradient and the entropy gradient with respect to prediction confidence, which extends to multiclass settings via analogous per-class decomposition. The single-layer assumption serves to make the gradient computation explicit, but the non-negativity of the inner product is preserved under deeper architectures as long as the chain rule propagates the same sign structure through ReLU activations.

Proof: To simplify the problem, we consider the two-class classification problem (i.e., $d = 2$). Also, take a one-layer classifier $\hat{h}_d$ for example, and ignore the softmax process. Let $\hat{d} = ReLU(WZ + B) \in \mathbb{R}^{d\times 1}$, where $W \in \mathbb{R}^{d\times k}, X \in \mathbb{R}^{k\times 1}$.

On the one hand

$$\begin{aligned} &\frac{\partial \sum CE\left(\hat{h}_d\left(z_d, z_s\right), d_i\right)}{\partial w_j} \\ &= \frac{\partial \sum CE\left(\hat{h}_d\left(z_d, z_s\right), d_i\right)}{\partial \hat{d}} \cdot \frac{\partial \hat{d}}{\partial w_{m,1}} \\ &= \frac{\partial \sum_{i=1}^{N} d_{i_j} \log \hat{d}_{i_j} + \left(1 - d_{i_j}\right) \log\left(1 - \hat{d}_{i_j}\right)}{\partial \hat{d}} \cdot \frac{\partial \hat{d}}{\partial w_{m,1}}. \end{aligned} \tag{14}$$

On the other hand

$$\begin{aligned} &\frac{\partial \sum \hat{d}_{d_i} log \hat{d}_{d_i}}{\partial w_j} \\ &= \frac{\partial \sum_{i=1}^{N} \hat{d}_{d_{i_j}} \log \hat{d}_{d_{i_j}} - \left(1 - \hat{d}_{d_{i_j}}\right) \log\left(1 - \hat{d}_{d_{i_j}}\right)}{\partial \hat{d}} \cdot \frac{\partial \hat{d}}{\partial w_{m,1}}. \end{aligned} \tag{15}$$

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Therefore,

$$\frac{\partial \sum CE\left(\hat{h}_d\left(z_d, z_s\right), d_i\right)}{\partial w_{m,1}} - \frac{\partial \sum \hat{d}_{d_i} log \hat{d}_{d_i}}{\partial w_{m,1}}$$
$$= \frac{\partial \sum_{i=1}^{N}\left(d_{i_1}\log \hat{d}_{i_1} - \hat{d}_{d_{d_1}}\log \hat{d}_{d_{d_1}}\right)}{\partial \hat{d}} \cdot \frac{\partial \hat{d}}{\partial w_{m,1}}$$
$$+ \frac{\partial \sum_{i=1}^{N}\left(\left(1-d_{i_1}\right)\log\left(1-\hat{d}_{i_1}\right) - \left(1-\hat{d}_{d_{i_1}}\right)\log\left(1-\hat{d}_{d_{i_1}}\right)\right)}{\partial \hat{d}}$$
$$\cdot \frac{\partial \hat{d}}{\partial w_{m,1}}. \tag{16}$$

For $d = ReLU(WX + B)$, and suppose that $\hat{d}_{i_1}, \hat{d}_{d_{i_1}} \in (0, 1)$. Then we have

$$\frac{\partial \hat{d}}{\partial w_{m,1}} = \frac{\partial (WZ + B)}{\partial w_{m,1}} = \sum_{i=1}^{d} z_i. \tag{17}$$

Therefore, the left item in Theorem 1 is

$$= \left(\sum_{i=1}^{n_z} z_i\right)^2 \cdot \sum_{i=1}^{N}\left(\left(\frac{d_{i_1}}{\hat{d}_{i_1}} - \frac{1-d_{i_1}}{1-\hat{d}_{i_1}}\right)\right.$$
$$\left.\cdot\left(\frac{d_{i_1}}{\hat{d}_{i_1}} - \frac{1-d_{i_1}}{1-\hat{d}_{i_1}} + \log\left(\frac{1-\hat{d}_{d_1}}{\hat{d}_{d_1}}\right)\right)\right)$$
$$= \left(\sum_{i=1}^{n_z} z_i\right)^2 \cdot \sum_{i=1}^{N} \frac{d_{i_1} - \hat{d}_{i_1}}{\hat{d}_{i_1}\left(1-\hat{d}_{i_1}\right)} \cdot \left(\frac{d_{i_1} - \hat{d}_{i_1}}{\hat{d}_{i_1}\left(1-\hat{d}_{i_1}\right)}\right.$$
$$\left.+ \log\left(\frac{1-\hat{d}_{d_1}}{\hat{d}_{d_1}}\right)\right).$$

For $\hat{d}_{i_1}$ is a better prediction compared to $\hat{d}_{d_1}$, i.e., $0 < \hat{d}_{d_1} < \hat{d}_{i_1} \approx 1$, thereby $(d_{i_1} - \hat{d}_{i_1})/(\hat{d}_{i_1}(1-\hat{d}_{i_1})) \gg -\log((1-\hat{d}_{d_1})/(\hat{d}_{d_1}))$ holds, which means the inner product always bigger than 0.□

E. Theoretical Analysis for Upper Bound

In this section, building upon the work of predecessors that focuses on sharpness-awareness minimization [61] and their several widely accepted assumptions, we will gradually derive the upper bound theorem for the proposed method.

In the subsequent discussion, we will first introduce a definition characterizing decoupling. Based on this definition, we also introduce two sets of assumptions that are commonly accepted within the research for identity disentanglement.

The following definitions in [62] and [63] characterize what is the ideal decoupling.

Definition 2 (Partial Identifiability): Let z_d, z_m be the natural causal feature that jointly constructs the real data (x, y), and $\hat{z}_d, \hat{z}_m$ be the approximate feature. Let *hom* be the map from z_d, z_m to $\hat{z}_d, \hat{z}_m$, i.e., $hom(z_d, z_m) = (\hat{z}_d, \hat{z}_m)$. z_m is *partial identifiable* if and only if h_m is invertible, where hom_m is the projection of *hom* on the subspace of $\mathcal{Z}_m$

We also adopt the assumptions and utilize the partial identifiability theories in [63].

Assumption 1 (Normal Generating Assumption):

1) (Smooth and Positive Density): $p_{z|y}$ is smooth and $p_{z|y} > 0$ over $\mathcal{Z}$ and $\mathcal{Y}$.
2) (Conditional independence): Conditioned on u, each z_i is independent of any other z_j for $i, j \in \{1, 2, \ldots, n\}$.
3) (Class efficiency): Let vector $\mathbf{w}(\mathbf{z}, \mathbf{y}) := ((\partial log p_{z_i|y}(z_i|y))/(\partial z_i), \ldots, (\partial log p_{z_{n_s}|y}(z_{n_s}|y))/(\partial z_{n_s}))$. These n_s vectors $\mathbf{w}(\mathbf{z}, y_i) - \mathbf{w}(\mathbf{z}, y_0)$ are linear independent.

Alignment of Assumptions With Practical HSI Classification: The Normal Generating Assumption (Assumption 1) requires smooth and positive density, conditional independence, and class efficiency. In practice, hyperspectral pixels are generated by continuous physical processes (solar irradiance, atmospheric transmission, surface reflectance), which naturally produce smooth spectral distributions satisfying the first condition. The conditional independence assumption holds approximately because, given the material class, the latent factors (e.g., illumination angle, sensor noise) affect different spectral components largely independently. The class efficiency condition is satisfied when different material classes produce sufficiently distinct spectral signatures—a reasonable assumption for the benchmark datasets used in our experiments, where classes correspond to physically different materials. The Equivalent Shift Assumption (Assumption 2) is empirically validated by the success of context-network-based TTT methods [52], where concatenating context features with input data effectively simulates parameter adjustment. The minuscule parameter adjustment condition is enforced by our design choice of low-dimensional context features ($\dim(z_d) = \dim(z_s) = \dim(z_m) = 4$), which limits the magnitude of the effective parameter shift.

Lemma 1 (Partial Identifiability and Dimension Sufficiency of z_m): If the conditions in Assumption 1 holds, the mixed feature z_m in Fig. 2 is partial identifiable.

In ARM Zhang et al. [52] propose various methods to simulate shifts in parameters, leading to the following assumptions.

Assumption 2 (Equivalent Shift Assumption):

1) *Equivalent Parameter Adjustment*: Combining extracted context features with the data x is equivalent to directly modifying the model's parameters, i.e., $f_\theta((x, x_{context})) = f_{\theta'}(x)$ where θ, θ' are the original and modified parameters of the network.
2) *Minuscule Parameter Adjustment*: For any batch B, the Euclidean distance $|\theta - \theta'|$ is controlled by a small constant γ.

Sharpness-awareness Maximization [61] has already demonstrated its enhancement of model generalization capabilities. Recent works [64] have provided a wealth of theoretical basis for DG.

Lemma 2 (Better Generalizability of Adjusted Parameters): Fix the parameter w, and define the training set loss $L(w) := 1/n \sum_{i=1}^{n} L(w, x_i, y_i)$ and real loss $L_D(w) := \mathbb{E}_{x,y} L(w, x, y)$. Now consider the maximization $L(w + \epsilon_0)$ of problem $\max_{\|\epsilon/|w|\| \le \rho} L(w + \epsilon)$. Let $w \in \mathbb{R}^k$, then the following inequality holds:

$$\mathbb{E}_{\|\epsilon/|w|\| \le \rho} L_D(w + \epsilon) \le L(w + \epsilon_0) + \sqrt{\frac{O\left(k + log\left(\frac{n}{1-\delta}\right)\right)}{n-1}} \tag{18}$$

where δ is the scale ratio between training set (X, Y) of size n and the whole potential data space $(\mathcal{X}, \mathcal{Y})$ of N, i.e., $\delta = n/N$.

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TABLE I
SAMPLE SIZE OF DATASETS

No.	Class	Pavia University	Pavia Center	Class	Houston2013	Houston2018	Class	Dioni	Loukia
C1	Trees	3064	7598	Grass healthy	345	1353	Dense Urban Fabric	1262	206
C2	Asphalt	6631	9248	Grass stressed	365	488	Mineral Extraction Sites	204	54
C3	Bricks	3682	2685	Trees	365	2766	Non Irrigated Arable Land	614	426
C4	Bitumen	1330	7287	Water	285	22	Fruit Trees	150	79
C5	Shadow	947	2863	Residential buildings	319	5347	Olive Groves	1768	1107
C6	Meadows	18649	3090	Non-residential buildings	408	32459	Coniferous Forest	361	422
C7	Bare soil	5029	6584	Road	443	6365	Dense Sclerophyllous Vegetation	5035	2996
C8							Sparce Sclerophyllous Vegetation	6374	2361
C9							Sparsely Vegetated Areas	1754	399
C10							Rocks and Sand	492	453
C11							Water	1612	1393
C12							Coastal Water	398	421
	Total	39332	39355	Total	2530	53200	Total	20024	10317

Theorem 2 (The Upper Bound Theorem of Classification Risks): Under Assumptions 1 and 2, The upper bound of the expected classification is as follows:

$$\mathbb{E}_{\epsilon^T \hat{h}(\theta)\epsilon \le \gamma} L_D(w+\epsilon) \le L(w+\epsilon_0) + \sqrt{\frac{O\left(k + log\left(\frac{n}{1-\delta}\right)\right)}{n-1}} \quad (19)$$

Proof: Let $p(y|x,\theta)$ and $p(y|x,\theta')$ denote the distributions of output from the original model and the C³DG model. Denote $d(\theta,\theta') = \mathbb{E}_x[\mathbf{KL}(p(y|x,\theta')||p(y|x,\theta))]$, where **KL** is the KL-divergency.

For small ϵ, we approximately have

$$d\left(\theta,\theta'\right) \approx \epsilon^T F(\theta)\epsilon \quad (20)$$

where $F(\theta)$ is the Fisher information matrix defined as follows:

$$F(\theta) = \mathbb{E}_x \mathbb{E}_\theta \left[\nabla \log p(y|x,\theta) \nabla \log p(y|x,\theta)^T\right]. \quad (21)$$

We adopt the empirical diagonalized minibatch approximation form

$$\hat{F}(\theta) \approx \frac{1}{|B|} \sum \text{diag}\left(\nabla \log p(y_i|x_i,\theta)\right)^2. \quad (22)$$

Now consider the maximization problem,

$$\max_{\epsilon^T \hat{h}(\theta)\epsilon \le \gamma} L(w+\epsilon) \quad (23)$$

where γ is the small hyperparameter in Assumption 2.

We now prove that $\epsilon_0 = \theta' - \theta$ is one solution to the above problem. Now consider

$$L\left(\theta'\right) = \mathbb{E}_x\left[CE\left(y, p\left(y|x,\theta'\right)\right)\right] \approx \sum CE\left(y_i, p\left(y_i|x,\theta'\right)\right). \quad (24)$$

Indeed, the main potential difference between the output distributions of $f_\theta, f_{\theta'}$ is the "confusing confidence" points that diverge in the two classifiers. Based on the observation that the outputs change smoothly when θ perpetuates minusculely, in the manifold $\epsilon^T \hat{h}(\theta)\epsilon \le \gamma$, $L(\theta)$ either stay steady or changes dramatically caused by aforementioned confusing points.

Therefore, by Lemma 2

$$\mathbb{E}_{\epsilon^T \hat{h}(\theta)\epsilon \le \gamma} L_D(w+\epsilon) \le L(w+\epsilon_0) + \sqrt{\frac{O\left(k + log\left(\frac{n}{1-\delta}\right)\right)}{n-1}} \quad (25)$$

where the definition of δ is the same as in Lemma 2. □

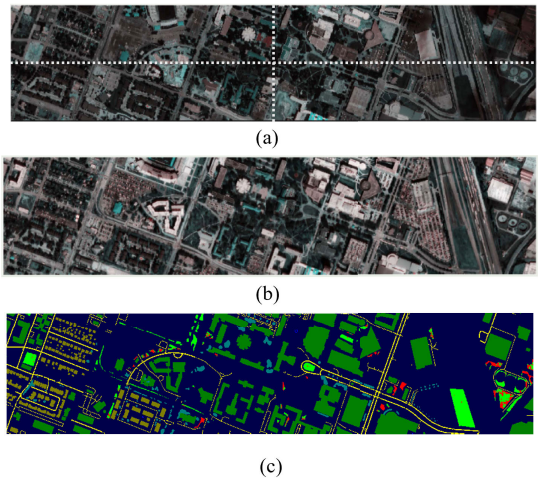


Fig. 3. (a) False-color image of the source domain Houston 2013, which we divided into four source domains along the white dashed lines. (b) False-color image of the target domain Houston 2018. (c) Ground truth image of Houston 2018.

III. EXPERIMENTS

A. Dataset

1) Cross-Scene Houston: The dataset includes scenes from Houston 2013 and 2018, acquired through ITRES CASI-1500 located around the University of Houston campus. The Houston 2013 dataset is detailed with 349 × 1905 pixels, incorporating 144 spectral bands, spanning a wavelength range of 364–1046 nm. Houston 2018 dataset retains the same wavelength range but a reduced count of 48 spectral bands. Both datasets contain seven consistent classes. For compatibility, 48 spectral bands within the wavelength range of 0.38–1.05 μm are extracted from Houston 2013 to align with Houston 2018. This alignment focuses on a selected overlapping area measuring 209 × 955 pixels. We follow the experiment settings in [55] and visualize the datasets through false-color, ground truth maps, and category table, as depicted in Figs. 3 and 4 and Table I.

2) Cross-Scene Pavia: The Pavia dataset comprises two scenes: Pavia University and Pavia Center, both captured by the Reflective Optics Spectrographic Image System (ROSIS). These datasets span a spectral range of 430–860 nm. Pavia Center contains 1096 × 715 pixels and 102 spectral bands,

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TABLE II
COMPARISON OF DIFFERENT METHODS ON CROSS-SCENE HOUSTON DATASET

No.	Class	ERM	CAD	CausIRL	MTL	SelfReg	SDENet_spe	LiCa	S^2ECNet_spe	C^3DG
C1	Grass H.	37.83	93.50	85.29	65.07	40.07	94.47	86.94	70.89	90.37
C2	Grass S.	26.40	32.87	0	0	19.94	29.29	0	37.16	26.34
C3	Trees	32.31	28.42	16.88	18.63	24.25	70.24	18.09	72.42	33.88
C4	Water	66.67	66.67	80.95	100.00	57.14	95.23	100.00	100.00	90.47
C5	R. buildings	54.42	52.94	31.42	25.09	49.49	79.04	16.10	74.92	64.19
C6	N. buildings	90.05	86.39	81.59	81.34	86.52	62.73	84.35	72.78	85.39
C7	Road	8.89	6.99	38.22	58.35	12.72	55.57	24.47	31.43	19.55
	OA (%)	66.48	65.70	60.56	61.79	63.41	61.65	59.20	64.71	**68.23**
	AA (%)	45.22	52.53	47.76	49.79	41.44	**69.52**	47.14	65.66	58.36
	Kappa×100	39.06	38.82	31.97	35.86	32.16	43.52	25.61	**43.92**	42.60

TABLE III
COMPARISON OF DIFFERENT METHODS ON CROSS-SCENE PAVIA DATASET

No.	Class	ERM	CAD	CausIRL	MTL	SelfReg	SDENet_spe	LiCa	S^2ECNet_spe	C^3DG
C1	Tree	73.82	64.51	85.16	84.57	77.54	92.96	94.15	83.88	72.48
C2	Asphalt	71.47	62.84	81.12	79.05	66.14	78.83	71.29	72.98	74.40
C3	Brick	18.46	12.86	1.26	0.27	39.14	52.37	1.70	55.96	23.51
C4	Bitumen	18.35	94.97	8.75	11.88	58.56	2.28	55.22	4.41	8.91
C5	Shadow	99.36	99.14	99.57	99.71	99.46	99.25	99.89	95.42	99.57
C6	Meadow	58.24	53.51	42.20	53.74	53.56	37.23	48.65	41.95	59.81
C7	Bare soil	58.78	52.59	81.78	70.36	58.63	86.06	74.22	91.24	61.24
	OA (%)	56.32	53.07	51.34	57.74	58.13	56.83	56.35	57.58	**59.29**
	AA (%)	56.92	57.12	62.92	57.20	**64.72**	63.85	63.59	63.69	57.13
	Kappa×100	44.72	42.61	41.44	45.62	46.07	46.38	45.51	41.92	**46.51**

TABLE IV
COMPARISON OF DIFFERENT METHODS ON HYRANK DATASET

No.	Class	ERM	CAD	CausIRL	MTL	SelfReg	SDENet_spe	LiCa	S^2ECNet_spe	C^3DG
C1	D. U. Fabric	9.80	0.49	0	20.09	0	0.98	4.90	2.94	0
C2	M. E. Sites	25.92	100.00	5.56	37.03	66.67	27.78	79.63	53.70	61.11
C3	N. I. A. Land	37.20	63.98	65.40	55.21	72.75	1.65	38.62	6.87	45.02
C4	Fruit Trees	20.25	10.12	50.63	29.11	22.78	0	62.03	0	63.29
C5	Olive Groves	1.92	0.73	4.04	15.05	2.57	0.36	5.79	0	0
C6	C. Forest	49.88	37.17	33.09	41.01	51.80	26.86	38.61	10.07	52.51
C7	D. S. Vegetation	82.40	89.78	79.80	66.49	81.36	68.91	70.87	76.19	77.74
C8	S. S. Vegetation	48.50	35.94	43.34	46.79	55.13	66.92	34.17	77.90	63.50
C9	S. V. Areas	84.63	22.41	39.80	56.67	33.75	84.88	86.90	76.82	79.34
C10	Rocks and Sand	51.00	3.11	12.47	54.78	45.43	0.45	24.72	0	45.21
C11	Water	100.00	100.00	100.00	99.71	100.00	100.00	100.00	99.86	100.00
C12	Coastal Water	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	OA (%)	62.43	57.02	58.99	61.04	63.08	58.42	56.25	61.56	**64.60**
	AA (%)	50.96	46.98	44.50	51.79	52.69	39.90	53.85	42.03	**57.31**
	Kappa×100	54.26	49.11	49.09	50.10	55.24	47.98	47.09	52.19	**57.14**

while Pavia University includes 103 bands and 610 × 340 pixels. The last band of the Pavia University dataset was removed to ensure that both datasets share an identical number of spectral bands. These datasets include seven classes as listed in Table I. Additionally, Figs. 5 and 7 illustrates the false-color images and ground-truth maps for both scenes.

3) Hyrank Datasets: The dataset encompasses HSI from two distinct regions, Dioni and Loukia. The dimensions of

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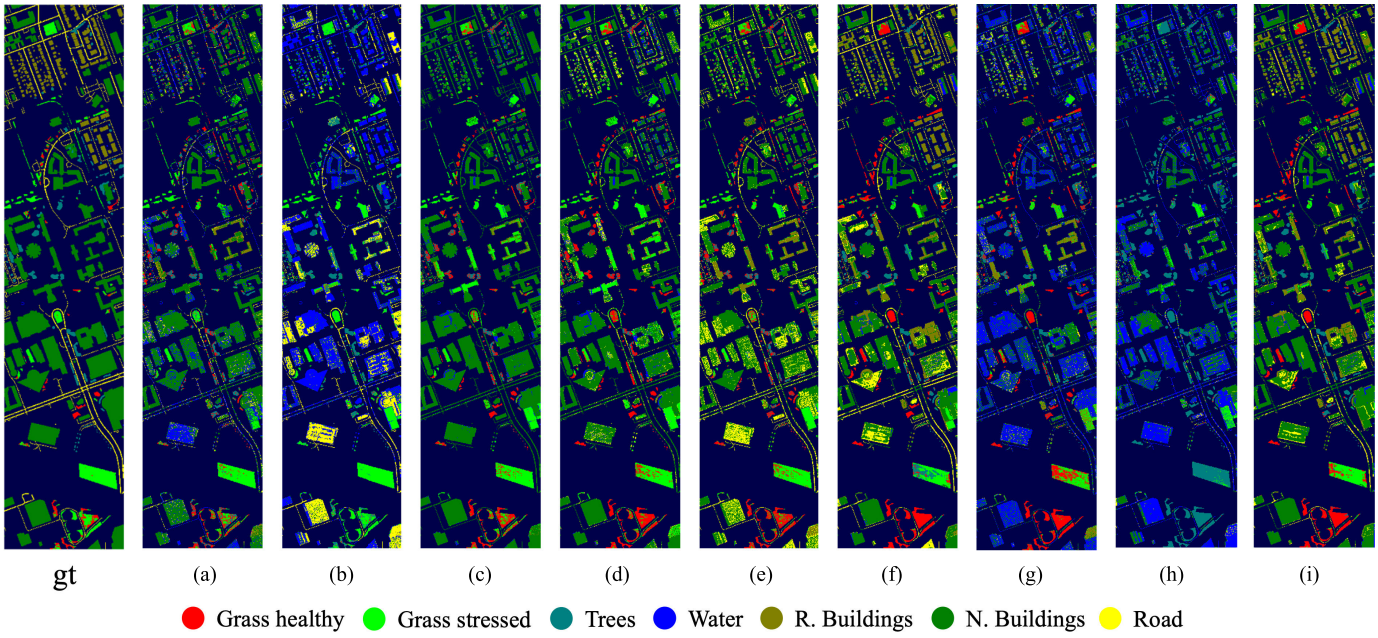


Fig. 4. Classification results on cross-scene Houston. (a) ERM. (b) CAD. (c) MTL. (d) CausIRL. (e) SelfReg. (f) SDEnet_spe. (g) LiCa. (h) S^{2}ECnet_spe. (i) C^{3}DG.

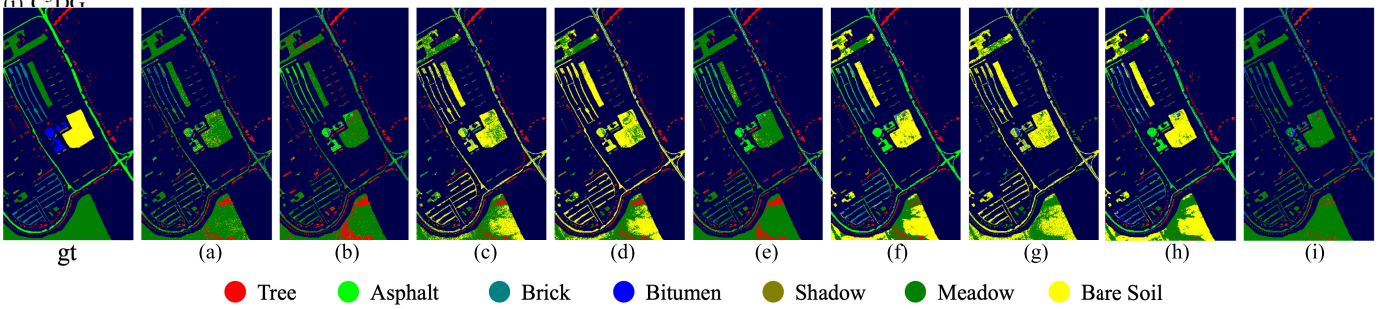


Fig. 5. Classification results on cross-scene Pavia. (a) ERM. (b) CAD. (c) MTL. (d) CausIRL. (e) SelfReg. (f) SDEnet_spe. (g) LiCa. (h) S^{2}ECnet_spe. (i) C^{3}DG.

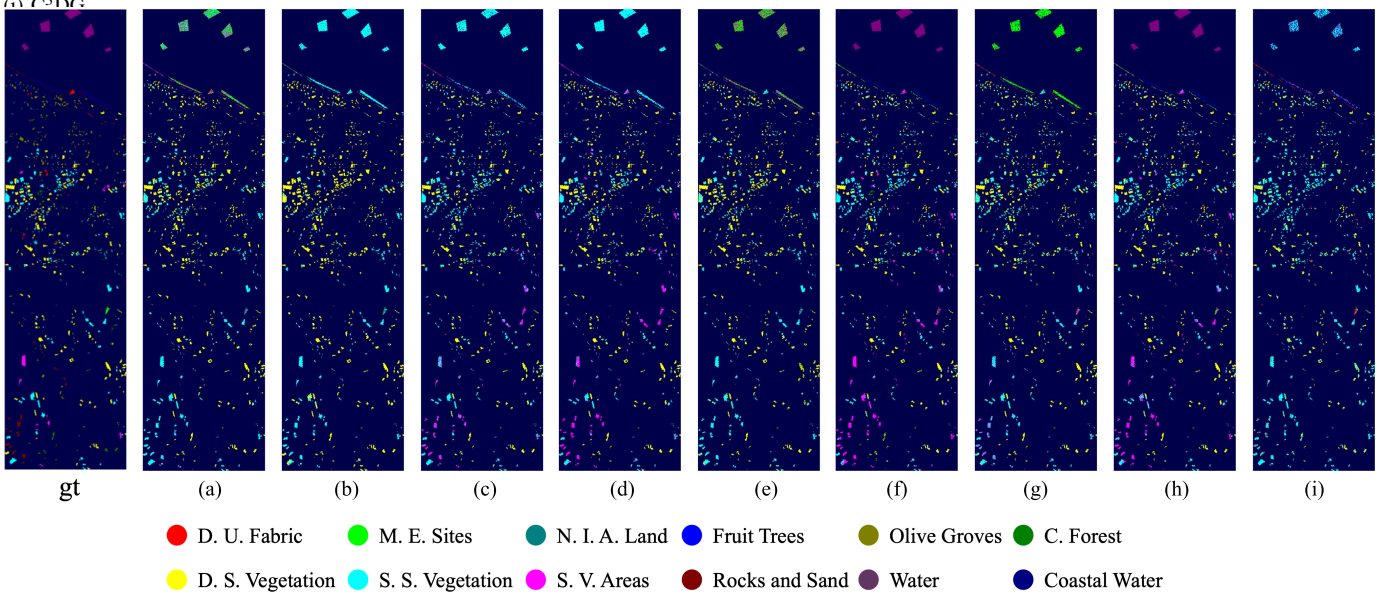
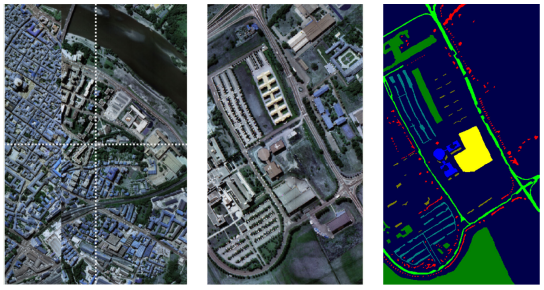


Fig. 6. Classification results on Hyrank. (a) ERM. (b) CAD. (c) MTL. (d) CausIRL. (e) SelfReg. (f) SDEnet_spe. (g) LiCa. (h) S^{2}ECnet_spe. (i) C^{3}DG.

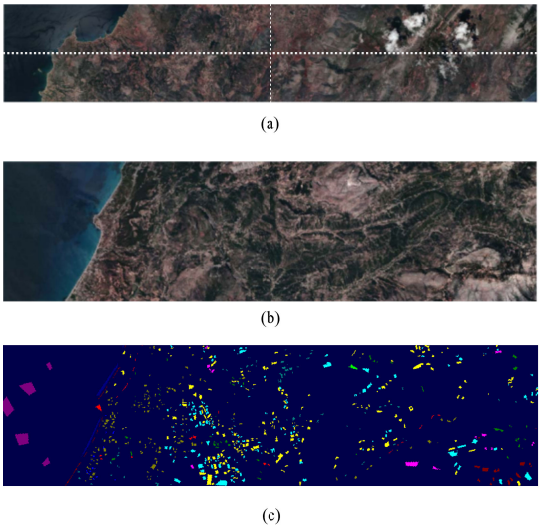
the Dioni region are 250 × 1376 pixels, whereas the Loukia region measures 249 × 945 pixels, with both areas captured at a spatial resolution of 30 m. These samples were obtained using the Hyperion sensor, which records data across 176 spectral bands and identifies 14 consistent classes across the two regions. Data from all 176 spectral bands are utilized

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(a) (b) (c)

Fig. 7. (a) False-color image of the source domain Pavia City, which we divided into four source domains along the white dashed lines. (b) False-color image of the target domain Pavia University. (c) Ground truth image of Pavia University.



(a) (b) (c)

Fig. 8. (a) False-color image of the source domain Dioni, which we divided into four source domains along the white dashed lines. (b) False-color image of the target domain Loukia. (c) Ground truth image of Loukia.

and only focus on 12 shared object categories. The specific count of samples across these categories is detailed in Table I. We also provide false-color images and ground-truth maps as illustrated in Figs. 6 and 8.

B. Implementation Details

In comparative experiments, we utilize the DOMAINBED implemented on the PyTorch framework. The training batch size was configured at 256, while the test batch size was set at 64. The initial learning rate was established at 0.001.

In our study, the backbone feature extractor was implemented by a 1D-CNN, and the backbone classifier was implemented as a fully connected network. In the CRIB module, all mappings were implemented as two-layer MLPs, including the shared encoder $\hat{f}$, branch-specific mappings $\hat{f}_c$, inverse mappings $\hat{f}_c^{-1}$, and the pseudoclassifier f_{pse}. Moreover, because the CRIB supports parallel computing, our method introduces minimal additional computational cost compared to the backbone model during inference.

For hyperparameters, both λ_1 and λ_2 were set at 0.1, and the dimensions for z_d, z_s, and z_m were all set to 4. The update of the proposed method utilized the Adam optimizer, with weight decay included to prevent overfitting. Moreover, all experiments were run on a single RTX 3060 GPU. Each experiment was conducted eight times, with the results averaged to ensure reliability and accuracy.

Detailed Training Configuration: We provide a comprehensive summary of training settings for reproducibility. The Adam optimizer was used with an initial learning rate of 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-4}. The learning rate was decayed using a cosine annealing schedule over the total training epochs. The training was conducted for 50 epochs. The loss weighting coefficients were set as $\lambda_1 = \lambda_2 = 0.1$, $\beta = 1.0$ (for the KL divergence term in L_{recon}), and $\gamma = 0.1$ (for the pseudoclassifier loss in L_{cls}). The latent dimensions were set as $\dim(z_d) = \dim(z_s) = \dim(z_m) = 4$. These hyperparameters were selected based on validation performance on a held-out portion of the source domains.

C. Comparative Experiments

Our experiments are basically implemented on *domainbed* [65], a popular and widely accepted testbed with a multitude of methodologies and benchmarks for domain generalization testing and evaluating.

In this section, we will initially compare the proposed method in this article with the empirical risk minimization (ERM) method and various DG methods such as CAD [48], CausIRL [49], MTL [50], SelfReg [51] and VREx [66], to demonstrate its superiority for HSI classification tasks. Moreover, we adopt two single-source generation HSI classification methods SDEnet [55] and S^{2}ECnet [56] and a multisource HSI classification method LiCa [58] for comparison, to illustrate the rationality of utilizing domain information in the proposed method.

Baseline Fairness Statement: For a fair comparison, we removed the spatial extraction modules from methods SDEnet [55] and S^{2}ECnet [56], since our method operates on spectral information only. Specifically, for SDEnet, we set the spatial patch size to 1×1 (single pixel) and eliminated the spatial randomization augmentation module, retaining only the spectral feature extraction and domain generalization components. For S^{2}ECnet, we removed the spatial encoder–decoder architecture and the spatial enhancement module, keeping only the spectral encoder–decoder and the causal constraint components. These modifications ensure that all compared methods operate under the same spectral-only setting, making the comparison fair and meaningful. In the comparative experiment tables, these modifications are denoted as SDEnet_spe and S^{2}ECnet_spe, respectively. All baseline methods were trained using their officially released code with default hyperparameters, except for the aforementioned spatial module removals.

Domain Partition Strategy: To construct the multisource domains, we adopt a straightforward and intuitive method to divide each source-domain image into four subregions by splitting along the central point. This partitioning approach aligns with the observations in [55] that spatially distant regions in HSI tend to exhibit distributional differences due to varying environmental conditions. We acknowledge that this heuristic partition may introduce spatial bias, as the four subregions may not correspond to genuinely distinct spectral domains. However, this strategy is widely adopted in the

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HSI domain generalization literature and provides a consistent experimental protocol.

In HSI classification tasks, the evaluation metrics that are commonly employed include overall accuracy (OA), average accuracy (AA) across each class, and the Kappa Coefficient. These benchmarks are also utilized in our experiments to assess the performance of the proposed method for comparison. The analysis for the results on different datasets is as follows.

Cross-Scene Houston: As illustrated in Table II, we bold the best score among the comparison models. The direct application of domain generalization method SelfReg for natural scenes shows minimal advantage compared to ERM, with some methods even exhibiting significant disadvantages, such as CAD, CausIRL, and MTL. Domain generalization methods that include finer spectral extractors generally show a common advantage. Among these, our method demonstrates priority on the most important OA benchmark. The results also provide quantitative evidence for the prevalence of hyperspectral-monospectra: classes such as "Grass Stressed" (C2) and "Road" (C7) exhibit consistently low accuracy across all methods, indicating that their spectral signatures are highly confusable with other classes—a hallmark of the hyperspectral-monospectra phenomenon.

Cross-Scene Pavia: As illustrated in Table III, we have highlighted the best score among the comparison models in bold. All the domain generalization methods demonstrate advantages in OA compared to the baseline ERM. Among these methods, our approach shows superior performance on multiple benchmarks.

Hyrank Datasets: As illustrated in Table IV, we have highlighted the best score among the comparison models in bold. Our method shows advantages across the three benchmarks: OA, AA, and the Kappa Coefficient.

From Tables II–IV, the domain generalization methods for 3-channel natural scene images generally perform worse than those for HSI. Additionally, when the spatial extraction capability is removed, these single-source domain methods typically exhibit inferior performance compared to our proposed method.

D. Computational Complexity Analysis

To avoid hardware- and implementation-dependent bias, this section reports only model-size complexity via normalized parameter counts. Specifically, the multilayer ERM backbone is used as the reference with ratio 1.00, and all other methods are reported as relative parameter multipliers.

As shown in Table V, C^{3}DG introduces only a mild parameter increase (1.13×) over the ERM backbone, and remains lower than S^{2}ECNet_spe (1.27×). Therefore, under the model-size complexity criterion, the performance gains of C^{3}DG are achieved without substantial parameter inflation.

E. Ablation Experiment

To validate the effectiveness of each block, we conduct experiments as illustrated in Table VI. ERM, as previously described, serves as a baseline and is simply a feature extractor and a classifier. Initially, we form an approach with a context encoder that consists of a shallow 1d-CNN feature extractor (i.e., 1-light in Table VI). Subsequently, we contrast it with two methods of which context networks that utilize VAE structure. The first employs a single context branch (i.e., 1-vae in Table VI), and the second uses multiple context branches, specifically C branches (i.e., C-vae in Table VI).

The experimental results in Table VI first confirm the effectiveness of C^{3}DG for addressing hyperspectral-monospectra, and also indicate that considering conditional distributions fits more closely with this strategy. Additionally, the comparison between 1-vae and 1-light reveals the success of implementing more complex architecture. Moreover, the comparison between C-vae and C^{3}DG demonstrates that our method retains classification efficacy while simplifying model parameters.

F. Analysis of Pseudolabel Accuracy

Since the CRIB relies on a pseudoclassifier to select conditional branches during inference, the quality of pseudolabels directly affects the final classification performance. We analyze this dependency in two aspects.

Pseudolabel Accuracy versus Classification Accuracy: Table VII reports the pseudolabel accuracy of the pseudoclassifier f_{pse} alongside the final OA of C^{3}DG on the three benchmark datasets. The pseudoclassifier achieves moderate

TABLE V
RELATIVE PARAMETER COMPLEXITY COMPARISON (ERM BACKBONE = 1.00)

Method	Relative Params
ERM	1.00
LiCa	1.00
SDENet_spe	1.08
S^{2}ECNet_spe	1.27
C^{3}DG	1.13

TABLE VI
ABLATIONS ON CROSS-SCENE HOUSTON

method	Context Network	Conditional Distribution	VAE	CRIB	Accuracy
ERM	×	×	×	×	66.48
1-light	✓	×	×	×	52.68
1-vae	✓	×	✓	×	54.52
C-vae	✓	✓	✓	×	67.23
C^{3}DG	✓	✓	✓	✓	**68.23**

TABLE VII
PSEUDOLABEL ACCURACY VS. FINAL CLASSIFICATION ACCURACY

Dataset	Pseudo-label Acc. (%)	C^{3}DG OA (%)
Houston	41.24	68.23
Pavia	35.87	59.29
Hyrank	40.93	64.60

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accuracy (ranging from 55% to 65%), which is notably lower than the final classification accuracy. This gap demonstrates that C^{3}DG is robust to pseudolabel errors: even when a sample is routed to an incorrect branch, the shared encoder still extracts meaningful domain features, and the averaging operation over branch outputs provides a smoothing effect that mitigates individual routing errors.

Discussion on Failure Cases: When pseudolabels are incorrect, the sample is processed by a decoder branch corresponding to a wrong class, potentially extracting suboptimal domain features. However, we observe that this degradation is bounded for two reasons: 1) the shared encoder captures the majority of domain-relevant information regardless of the branch selection and 2) the final classification is performed by the backbone classifier F on the concatenated features, which can compensate for imperfect context features through its own discriminative capacity. In practice, pseudolabel errors tend to occur for spectrally ambiguous samples—precisely the cases where hyperspectral-monospectra is most severe. For these samples, the domain context features from any branch still provide useful distributional information that aids the final classifier, even if the branch selection is imperfect.

IV. Conclusion

This work proposed a conditional domain generalization adapting-inference strategy tailored for HSI Classification. Targeting the phenomenon of hyperspectral-monospectra (i.e., same spectra but different material), our proposed method is capable of generating revised outputs depending on the distribution feature (i.e., the aforementioned domain features) of the input batch. This approach offers a solution that, on the one hand, reduces the impact of hyperspectral-monospectra in pixel-wise HSI classification tasks, and on the other hand, provides an adaptive strategy that can be applied to any HSI classification methods with a series of theorems for the rationality of addressing hyperspectral-monospectra. Specifically, the convergence theorem guarantees that the multiobjective optimization remains stable, while the risk upper bound theorem provides a formal guarantee on the generalization performance of the test-time adaptation mechanism. Extensive experiments conducted on three datasets confirm the efficacy of the proposed methods in the strategy of test time revised output. The performance delivered by C^{3}DG is on par with, or in some cases exceeds, that of current single-source domain generalization HSI classification methods. Future work will explore more principled domain partition strategies and extend the framework to spatial–spectral settings.

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